

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.

Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

The goal of this observing program is to obtain ground-based photometric observations of exoplanet transits in order to measure and characterize their transit light curves. The resulting light curves will be analyzed using differential photometry, comparing the brightness of the target star with nearby comparison stars to remove changes in atmospheric transparency and other observational effects. From the shape of each transit, we can measure quantities such as the transit depth, duration, and mid-transit time. These measurements can be compared with previously published values and predicted ephemerides. Repeated transit observations are particularly useful for improving the orbital ephemerides of exoplanets. Comparing our measured transit times with predicted times can reveal deviations that may indicate transit-timing variations or inaccuracies in the currently known orbital period. The observations will therefore contribute independent measurements of well-studied planetary systems while providing practical constraints on their orbital and transit parameters. This program is interesting because relatively small ground-based telescopes can make precise measurements of planetary transits, allowing us to independently verify and refine parameters obtained from larger surveys and observatories. The observations will also provide practical experience with time-series astronomical photometry and the analysis of exoplanet transit light curves.

Telescope Requested (0.5 points): Mt Bigelow 61" Kuiper Telescope

Instrument (0.5 points): Mont 4k

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

A dark moon is preferred, but a grey Moon is acceptable for this observing program because the targets have relatively bright apparent magnitudes and relatively deep transits. Moonlight increases the sky background, which increases the amount of background noise in each photometric measurement and can reduce the signal-to-noise ratio of the target and comparison stars. This effect is particularly important for fainter targets and at shorter wavelengths, where scattered moonlight can contribute significantly to the background. The expected transit depths of approximately 1.5–1.75% are large enough that the transits should remain detectable under moderate sky-background levels.

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Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example “Object can be observed Feb-March.”

Object Name	Right Ascension	Declination	V magnitude	Object Type	Visibility
TOI-1516b	340.08462	69.50373	10.9	Exoplanet	10/2/26, 22:42-3:28
HD 209458b	330.79504	18.88424	7.7	Exoplanet	10/3/26, 20:21-1:27

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? **(2 points)**

The amount of time required for our program is around 3 hours, the exposure time should be around 5 seconds, and a signal to noise of around 1000 is necessary.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? **(2 points)**

The observations will allow us to obtain transit light curves for TOI-1516 b and/or HD 209458 b and measure their transit depths, durations, and mid-transit times. These measurements can be compared with published predictions to refine the planets' orbital parameters and search for possible transit-timing variations. The observations will also provide independent ground-based measurements that can be compared with previous observations, helping verify and improve our understanding of these planetary systems.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. **(0.5 points)**

Archival TESS observations of TOI-1516 b will be analyzed to investigate the transit properties and orbital timing of this hot Jupiter. Publicly available TESS light curves will be obtained from the Mikulski Archive for Space Telescopes (MAST), and individual transit events will be identified and analyzed.